A BEGINNER PORTFOLIO GUIDE

15 AI/ML Projects for Beginners

Pick one project. Build a small version.
Show what worked and what you learned.



PREPARED BY

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CHOOSE A STARTING POINT

Try **House Price Predictor** for numbers, **Review Sentiment Checker** for text, or **Job Description Skill Finder** for a simple rules-based project.

Open the project index

Pick your first project

You don’t need to build all 15. Finish one small project you can explain, then choose a second if you want to try a different skill.

SET UP YOUR WORKSPACE

Open Google Colab and create a notebook. Run `print("Hello, AI!")` in the first cell. If your project uses a CSV, upload it using the Files sidebar and open it with `pandas.read_csv()`. Save the notebook to Drive or download it.

READ THE DIFFICULTY LABELS

Start here uses a small dataset or simple rules. **Core ML** introduces a few more steps. **Stretch** needs extra care with rare examples or images. Company names illustrate industry use cases.

PROJECT INDEX

01 House Price Predictor
02 Weekly Sales Forecaster
03 Customer Cancellation Predictor
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14 Job Description Skill Finder
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Portfolio checklist / Datasets and learning resources

House Price Predictor

START HERE / Real estate / Zillow’s Zestimate

THE PROBLEM

A homebuyer wants a rough price estimate. Use past housing data to predict the recorded median home value for an area.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

California Housing [D1]. Start with median income, house age and average rooms. The target is the area’s median value, recorded in units of $100,000.

BUILD YOUR FIRST VERSION

01. Load the data with fetch_california_housing(as_frame=True). Inspect the columns and plot median income against home value.

02. Set aside 20% of rows for testing with a fixed random seed. For a first comparison, predict the training median for every test row.

03. Fit LinearRegression on the training rows. Compare its average absolute error with the simple guess and convert the error to dollars.

04. Plot actual vs. predicted values and inspect five large misses. Explain that the data is historical and describes areas.

WHAT TO SHOW IN YOUR PORTFOLIO

A notebook with a scatter plot, two error scores and one example estimate with its limits.

INTERVIEW PRACTICE

How would you check if it works in a region the model has never seen?

Project index / Dataset and tutorial links

Weekly Sales Forecaster

START HERE / Retail / Walmart demand planning

THE PROBLEM

A shop needs to plan next week’s stock. Estimate how many units of one product it will sell next week.

TOOLS TO USE

Google Colab, pandas, matplotlib

DATA TO START WITH

UCI Online Retail [D2]. Pick one frequently sold StockCode. Use InvoiceDate and Quantity.

BUILD YOUR FIRST VERSION

01. Remove cancellations and nonpositive quantities. Sum units into complete calendar weeks for one item. Keep zero weeks only if you know the shop was operating.

02. Hold back the last eight complete weeks. Use last week’s sales as your first forecast.

03. Compare that with the average of the previous four weeks. For each test week, calculate the forecast using only data available before that week.

04. Plot both forecasts against actual sales. Compare the average absolute error in units and look for misses around spikes or promotions.

WHAT TO SHOW IN YOUR PORTFOLIO

A chart of eight weekly forecasts, a small error table and a recommendation for the shop owner.

INTERVIEW PRACTICE

What would you do if the product sold out and the recorded sales understated demand?

Project index / Dataset and tutorial links

Customer Cancellation Predictor

CORE ML / Telecom / Dialog Axiata

THE PROBLEM

A telecom team wants to prioritize customer outreach. Predict whether a customer in a sample dataset is marked as having left.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

IBM Telco Customer Churn [D3]. Use tenure, monthly charges and contract type. The answer column is Churn.

BUILD YOUR FIRST VERSION

01. Drop customerID and convert TotalCharges to numbers if you use it. Check blanks and count how many customers left.

02. Use a train/test split that keeps the share of cancellations similar. Build a simple guess that predicts everyone stays.

03. Use OneHotEncoder for contract type and LogisticRegression. Fit any cleaning and encoding on the training data only.

04. On the test set, count missed cancellations and false alerts. Report recall, the share of cancellations you caught, and precision, the share of alerts that were correct.

WHAT TO SHOW IN YOUR PORTFOLIO

A notebook with a comparison against the simple guess, a table of correct and wrong predictions, and a short outreach example.

INTERVIEW PRACTICE

What tradeoff would you make if the team could contact only 100 people?

Project index / Dataset and tutorial links

Review Sentiment Checker

START HERE / E-commerce / Amazon Review Highlights

THE PROBLEM

A store wants a quick view of feedback. Classify a short product review as positive or negative.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

UCI Sentiment Labelled Sentences [D4]. Start with the amazon_cells_labelled.txt file. Each line contains review text and a 0 or 1 label.

BUILD YOUR FIRST VERSION

01. Read the tab-separated file. Remove duplicate review text and inspect several examples of each label.

02. Split into training and test sets with similar label shares. Use the most common training label as a simple comparison.

03. Put TfidfVectorizer and LogisticRegression in a pipeline. The vectorizer turns words into numbers based on how common they are. Fit the pipeline on training reviews only.

04. Report the percentage classified correctly and read five mistakes. Write a few new reviews with mixed feelings or sarcasm and try them separately.

WHAT TO SHOW IN YOUR PORTFOLIO

A notebook that accepts a new review, prints a label and explains two kinds of mistakes.

INTERVIEW PRACTICE

How might your model behave on reviews from a different type of shop?

Project index / Dataset and tutorial links

Spam Message Detector

START HERE / Email and messaging / Google Gmail

THE PROBLEM

A messaging app needs to filter spam while keeping real messages visible. Label each SMS as spam or legitimate.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

UCI SMS Spam Collection [D5]. The SMSSpamCollection file has a label and message separated by a tab. “Ham” means legitimate.

BUILD YOUR FIRST VERSION

01. Read the file, remove duplicate messages and count the two classes. Keep the texts and labels in separate columns.

02. Split the data with similar class proportions. Start with the simple guess that all messages are legitimate.

03. Train a pipeline with TfidfVectorizer and MultinomialNB. Test it on the held-out messages.

04. Report spam precision and recall. Read real messages incorrectly marked as spam. Try made-up examples such as a legitimate discount notification.

WHAT TO SHOW IN YOUR PORTFOLIO

A notebook with a spam checker, a table of errors and three example messages for a demo.

INTERVIEW PRACTICE

Which mistake is more costly for this app and how would you reduce it?

Project index / Dataset and tutorial links

Fraud Alert System

STRETCH / Payments / Stripe Radar

THE PROBLEM

A payment team can review only a small number of transactions each day. Rank transactions by how suspicious they look.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

Simulated transactions [D6]. Use a small time slice and keep amount, transaction time and the fraud label.

BUILD YOUR FIRST VERSION

01. Sort by time and make three consecutive blocks for training, practice and final testing. Keep the natural share of fraud in each.

02. Start by ranking the biggest payments first. Then train a class-weighted LogisticRegression using amount and hour of day. Leave out fields created after a decision.

03. Use the middle block to compare both approaches at a fixed review limit, such as the top 20 alerts per day. Lock the approach before opening the final block.

04. On final test days, count how many frauds appear in the top 20 and how many alerts are false. Compare with the amount-only ranking.

WHAT TO SHOW IN YOUR PORTFOLIO

A daily alert shortlist, a comparison at the same review limit and a note that the transactions are simulated.

INTERVIEW PRACTICE

What happens to the results if the review team can handle only five alerts a day?

Project index / Dataset and tutorial links

Product Recommender

CORE ML / E-commerce / Amazon recommendations

THE PROBLEM

A shopper is viewing one item. Suggest three related products based on which items other shoppers bought together.

TOOLS TO USE

Google Colab, pandas

DATA TO START WITH

UCI Online Retail [D2]. Use InvoiceNo as the basket ID, StockCode as the item ID and Description as the display name.

BUILD YOUR FIRST VERSION

01. Remove cancellations, returns and obvious nonproduct codes. Collapse repeat lines for the same item in a basket. Use a small subset of complete baskets.

02. Set aside the latest baskets. In the earlier baskets, count how often each item pair appears together. Pick the three most frequent partners for an item.

03. Use the most popular training products as a fallback for an unseen item. Exclude the current item from all suggestions.

04. In each test basket with two or more products, pick one as the input. Check whether any of the three recommendations appears in the rest. Compare with recommending popular products.

WHAT TO SHOW IN YOUR PORTFOLIO

A notebook or small lookup app with five sample items and their recommendations, including one fallback example.

INTERVIEW PRACTICE

How would you recommend a new product with no purchase history?

Project index / Dataset and tutorial links

Delivery Time Predictor

CORE ML / Food delivery and logistics / DoorDash

THE PROBLEM

A takeaway shop wants a time estimate before a driver leaves. Predict the number of minutes between pickup and delivery.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

Make a clearly labeled practice CSV with distance_km, order_hour, weather, pickup_time and delivered_time. Use optional synthetic recipe [P1].

BUILD YOUR FIRST VERSION

01. Calculate minutes from the two timestamps. Check for missing or negative durations. Use delivered_time only to create the answer.

02. Hold back the most recent orders. Compare against a simple estimate using the median duration from earlier orders.

03. Train a small DecisionTreeRegressor using distance, order hour and weather. Encode weather using training data and limit the tree depth.

04. Report the average absolute error in minutes. Compare errors for short and long trips, and print a few predictions next to actual durations.

WHAT TO SHOW IN YOUR PORTFOLIO

A chart showing where estimates are off and a demo that takes order details and returns minutes.

INTERVIEW PRACTICE

Which details would you need to collect before trying this on real deliveries?

Project index / Dataset and tutorial links

Customer Group Finder

CORE ML / Retail marketing / Adobe Commerce audience tools

THE PROBLEM

A store wants to understand shopping habits. Group customers using how recently, how often and how much they bought.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

UCI Online Retail [D2]. Use CustomerID, InvoiceNo, InvoiceDate, Quantity and UnitPrice.

BUILD YOUR FIRST VERSION

01. Remove cancellations, nonpositive quantities or prices and rows with missing customer IDs. Calculate line totals from quantity times price.

02. Build one row per customer. Calculate days since the last purchase as of a fixed date, unique order count and total spend.

03. Standardize the three columns so one scale does not dominate. Try KMeans with three groups, then compare with four.

04. Describe each group using its typical values and size. Run with another random seed and check whether the broad patterns still make sense.

WHAT TO SHOW IN YOUR PORTFOLIO

A chart of the groups, a profile table and one practical marketing idea per group.

INTERVIEW PRACTICE

How would you explain the groups to a shop owner without using model terminology?

Project index / Dataset and tutorial links

Product Defect Detector

STRETCH / Manufacturing / BMW Group visual checks

THE PROBLEM

A factory worker wants a second check on items. Flag unusual photos of one type of product taken under consistent lighting.

TOOLS TO USE

Google Colab, OpenCV, NumPy

DATA TO START WITH

MVTec AD [D7]. Choose one category, such as bottles. Use the normal training images and the separate labeled test folders.

BUILD YOUR FIRST VERSION

01. Resize photos to the same size and inspect alignment. Start with one category and keep the original test folders untouched.

02. Create a reference image by averaging a subset of normal training photos. Score each photo using its average pixel difference from the reference.

03. Set a cutoff using remaining normal training photos, for example the 95th percentile of their scores. Freeze it and test on the supplied normal and defect images.

04. Count missed defects and false alerts. Show a few marked difference images and check whether lighting or alignment explains mistakes.

WHAT TO SHOW IN YOUR PORTFOLIO

A photo grid with predictions, a table of errors and a note about which defects this simple approach misses.

INTERVIEW PRACTICE

What would you change if the camera angle shifted? How would you check that it helped?

Project index / Dataset and tutorial links

Ask Your PDF

CORE ML / Education and document tools / Adobe Acrobat

THE PROBLEM

A student wants to find a fact in a handbook. Return the best matching passage and its page number for a typed question.

TOOLS TO USE

Google Colab, pypdf, scikit-learn

DATA TO START WITH

Make a short, text-based FAQ PDF or use a document you have permission to reuse. Write eight questions, including two with no answer.

BUILD YOUR FIRST VERSION

01. Extract text page by page. Split long pages into short passages and keep each passage's page number.

02. Use TfidfVectorizer and cosine similarity to rank passages by word overlap with the question. Return the top passage with its source page.

03. Try a score cutoff on practice questions and show "No supporting passage found" when the match is weak. Keep different questions for the final check.

04. Check whether each answerable question returns a passage with the right fact. Inspect the unanswered questions and test a question that uses different wording.

WHAT TO SHOW IN YOUR PORTFOLIO

A searchable notebook that shows the passage and page, plus a small table of test questions and outcomes.

INTERVIEW PRACTICE

When would a system that matches meaning help more than word matching?

Project index / Dataset and tutorial links

Support Ticket Sorter

START HERE / Customer support / Zendesk intelligent triage

THE PROBLEM

A help desk wants to send new messages to the right team. Label each request as billing, login or technical support.

TOOLS TO USE

Google Colab, pandas, scikit-learn

DATA TO START WITH

Write 20 varied fictional tickets per category. Use ticket_text and team columns. Include a few ambiguous tickets for separate manual checks.

BUILD YOUR FIRST VERSION

01. Write examples with different wording and lengths. Keep near-duplicate messages together so they cannot appear in both training and testing.

02. Train a TfidfVectorizer plus LogisticRegression pipeline. Compare it with a keyword rule such as “refund” or “invoice” for billing.

03. Check accuracy for each team. If you add a confidence cutoff, choose it on a small practice set and send low-confidence messages to human review.

04. On the separate test set, inspect wrong routes and report how many messages were sent for review. Demo three clear requests and one ambiguous one.

WHAT TO SHOW IN YOUR PORTFOLIO

A notebook that suggests a team, plus an error table and a manual-review option.

INTERVIEW PRACTICE

What should happen when a message contains both a billing and a login problem?

Project index / Dataset and tutorial links

Invoice Reader

CORE ML / Accounting / Microsoft AI Builder

THE PROBLEM

A small business wants to reduce manual entry. Extract the date, invoice number and total from a simple invoice.

TOOLS TO USE

Google Colab, pypdf, Python re, pandas

DATA TO START WITH

Create 12 fictional invoices as text PDFs. Use two layouts and manually record the correct fields in a spreadsheet.

BUILD YOUR FIRST VERSION

01. Extract text from the first layout with pypdf. Print the first few lines to see how the fields appear. Start with selectable text.

02. Write patterns that look for labels such as “Invoice Number” and “Grand Total.” Return “needs review” when a field is absent.

03. Test on unseen invoices from both layouts. Include an invoice with subtotal, tax and grand total, and one with a missing date.

04. Compare each extracted field against your answer sheet. Optionally try Tesseract on a scan and report those results separately.

WHAT TO SHOW IN YOUR PORTFOLIO

A CSV of extracted fields, a field-by-field score and two examples where review is needed.

INTERVIEW PRACTICE

How would you stop the program from mistaking a subtotal for the amount due?

Project index / Dataset and tutorial links

Job Description Skill Finder

START HERE / Recruiting / LinkedIn Skills Match

THE PROBLEM

A job seeker wants a short checklist from a posting. Pull out skill names and show the original line where each one appeared.

TOOLS TO USE

Google Colab, Python re, pandas

DATA TO START WITH

Write ten sample job descriptions. Make a small skill list with names and aliases, such as “Power BI” and “Microsoft Power BI.”

BUILD YOUR FIRST VERSION

01. Mark the skills by hand before running the tool. Include “required” and “nice to have” headings and some posts with no skill match.

02. Build case-insensitive phrase matching that respects word boundaries. Save the line of evidence and normalize aliases to one name.

03. Use explicit section headings to separate required and optional skills. Mark unknown when the posting gives no clear heading.

04. Use some posts to improve your rules and keep others for the final check. Count missed skills and incorrect matches, including short terms like R.

WHAT TO SHOW IN YOUR PORTFOLIO

A skills checklist with evidence lines, two sample posts and a brief explanation of what the rules miss.

INTERVIEW PRACTICE

How would you handle a skill your dictionary has never seen?

Project index / Dataset and tutorial links

Meeting Notes to To-Do List

START HERE / Team collaboration / Zoom AI Companion

THE PROBLEM

A project team wants to turn notes into a task list. Extract the task, owner and due date when those details are written in the notes.

TOOLS TO USE

Google Colab, Python re, pandas

DATA TO START WITH

Create 12 fictional meeting notes. Include clear assignments, missing dates and ordinary discussion lines with no task.

BUILD YOUR FIRST VERSION

01. Start with lines like “Action: Send the draft | Owner: Ana | Due: 2026-12-02.” Write rules to split each task into fields.

02. Allow missing owners and dates and mark them “not specified.” Attach the original line to each row so users can check it.

03. Use a few notes to refine the rules, then test on separate notes. Compare the task rows against a checklist you wrote by hand.

04. Show cases the rules missed, such as “Ana will send the draft on Friday.” Keep the simple parser as a comparison if you later try an AI model.

WHAT TO SHOW IN YOUR PORTFOLIO

A CSV with task, owner, due date and evidence, plus a count of missed tasks and wrong fields.

INTERVIEW PRACTICE

How would you handle “next Friday” when the meeting date is unknown?

Project index / Dataset and tutorial links

Finish and share your project

USE THIS NOTEBOOK STRUCTURE

1. Write the question you're trying to answer.
2. Load the data and show a few rows.
3. Explain what you cleaned and what you set aside for testing.
4. Build your simple first version and record the result.
5. Show examples of mistakes and what you'd change next.

WRITE A SHORT README

Include the problem, data source, instructions to run the notebook, a screenshot or chart, and your result. Use your actual numbers. Credit the dataset and check its terms before uploading any files.

TEST BEFORE SHARING

Restart the notebook and run it from top to bottom. Make sure it can find the data, regenerate the result, and show an example output. Keep test examples separate from the examples you used to make changes.

EXPLAIN YOUR PROJECT IN AN INTERVIEW

"I built this for [user] to solve [problem]. I compared my approach with [simple first guess] and measured [result] on data I held back. One thing it struggled with was [example]. My next step would be [specific improvement]."

STARTER VOCABULARY

A **feature** is an input such as distance. A **target** is the answer you want to predict. A **baseline** is a simple first guess you compare against. **Test data** is data you save to check your work after building it.

Datasets and starter examples

Tap a resource name to open its source.

D1 California Housing
Project 01. Load it directly with scikit-learn.

D2 UCI Online Retail
Projects 02, 07 and 09. Download the workbook and start with a small slice.

D3 IBM Telco Customer Churn
Project 03. Open the CSV and choose Download raw file.

D4 UCI Sentiment Labelled Sentences
Project 04. Unzip the files and start with the Amazon review file.

D5 UCI SMS Spam Collection
Project 05. Unzip the labeled message file.

D6 Fraud Detection Handbook sample data
Project 06. Follow the repository's loading instructions. Use several consecutive days per split and check each split contains fraud.

D7 MVTec AD images
Project 10. Use one category and check the dataset's usage terms before sharing images.

P1 / PRACTICE DELIVERY DATA

For project 08, make 150 fictional orders over 15 days. Vary distance from 1 to 12 km and add busy hours and rain. For a practice duration, use 8 + 3 × distance, add 5 minutes for rain and a few random minutes of noise. Add the duration to pickup_time to get delivered_time. Use the last 3 days for testing. The score describes this simulation.

Learn the tools as you build

Google Colab
Run Python in your browser and share your notebook.

pandas getting started
Read data, clean columns, group rows and work with dates.

scikit-learn getting started
Fit a model, combine steps in a pipeline and evaluate results.

Google Machine Learning Crash Course
Start with the regression, classification and data modules.

pypdf text extraction
Read text and page numbers from a PDF for projects 11 and 13.

Tesseract OCR
Optional for reading text from scanned invoices.

Streamlit getting started
Optional when you want to turn a working notebook into a simple app.

GitHub README guide
Document what you built, how to run it and what you learned.

Back to the project index